Latest version: https://vuphan314.github.io
This version: 2022 Oct 07

Vu H. N. Phan

Curriculum Vitae

Portland, Oregon
vuphan314.github.io
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Vu is a Formal Verification engineer at Intel Corporation, performing pre-silicon validation of a hardware design. He earned a Ph.D. in Computer Science at Rice University, advised by Prof. Moshe Vardi.

Experience

2022-now **Computer Engineer**, *Intel Corporation*
- Performing pre-silicon Formal Verification of an Intellectual Property in a System on a Chip.

2021 May-Aug **Technical Intern**, *Intel Corporation*
- Developed software to benchmark Formal Verification platforms.

Certification

2021 Jul **SVA, Formal, and JasperGold Fundamentals for Designers v19.03**, *Cadence Design Systems*

Languages

Hardware **SystemVerilog**
Software **Python, C++; Dafny, Coq**

Education

2017-2022 **Ph.D. in Computer Science**, *Rice University*, GPA 3.61/4.00
2017-2019 **M.S. in Computer Science**, *Rice University*, GPA 3.60/4.00
2014-2017 **B.S. in Computer Science & Math**, *Texas Tech University*, GPA 4.00/4.00

Projects

2020-2021 **ProCount**, *Projected Counter*, https://github.com/vardigroup/DPMC
- Implemented a weighted projected #SAT framework with Jeffrey Dudek.

2019-2020 **DPMC**, *Dynamic-Programming Model Counter*, https://github.com/vardigroup/DPMC
- Implemented a weighted #SAT framework with Jeffrey Dudek.

2018-2019 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, https://github.com/vardigroup/ADDMC
- Implemented a weighted #SAT solver.

2016-2018 **LED**, *Language of Effective Definitions*, https://vuphan314.github.io/LED
- Translated the literate-programming language LED into SequenceL and LaTeX.

2015-2016 **L**, *Logic*, https://github.com/iensen/LtoASPtranslator
- Translated the logic-programming language L into Answer Set Prolog with Evgenii Balai.

Coursework

2017-2021 **Graduate Level**

- Reasoning about Software
- Bioinformatics: Sequence Analysis
- Statistical Machine Learning
- Computer Systems Architecture
- Multi-Core Computing
- Compiler Construction
- Artificial Intelligence
- Programming Languages
- Logic in Computer Science

2014-2017 **Undergraduate Level**

- Operating Systems
- Database Systems
- Computer Architecture
- Programming Languages
- Software Engineering
- Algorithms
- Object-Oriented Programming
- Computer Organization and Assembly Language Programming
- Modern Digital System Design
- Data Structures
- Automata
- Programming Principles

Bibliography

Vu Phan is the correspondence author of the following publications (authors are sorted by surnames).

Theses

[Pha22] Vu H. N. Phan. "Quantitative Reasoning on Hybrid Formulas with Dynamic Programming". PhD thesis. Rice University, 2022. URL: https://scholarship.rice.edu/handle/1911/113243.

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: https://scholarship.rice.edu/handle/1911/107761.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: http://jeffreydudek.com/papers/sat21_procount.pdf.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: https://arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: https://arxiv.org/abs/1907.05000.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: https://arxiv.org/abs/1809.09319.